S1SBP6A EVK(NTF2432) User Guide

S.LSI New Business T/F

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1 Document Information

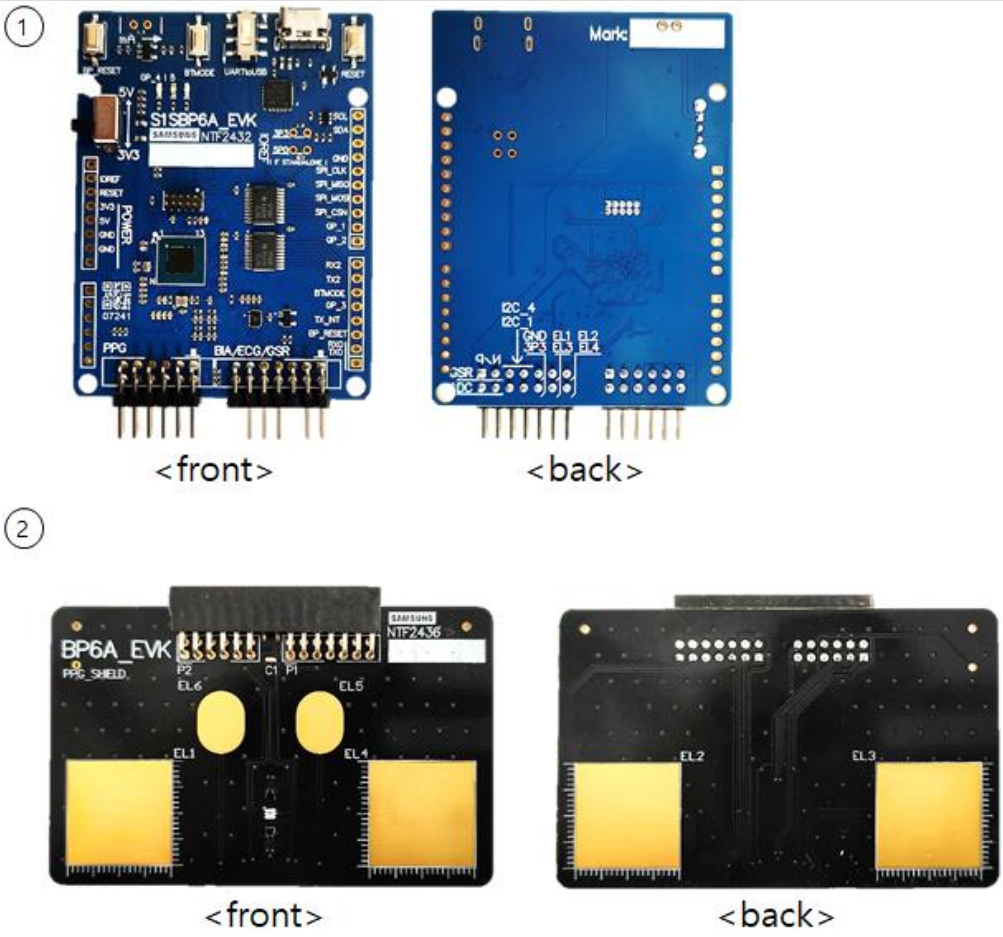
1.1 Revision History

Version	Date	Description	Author
1.0	2024-09-11	Initial Draft	Wonhyuk Jung (whyuk.jung@samsung.com)

2 Hardware User Manual (BP6A EVK NTF2432)

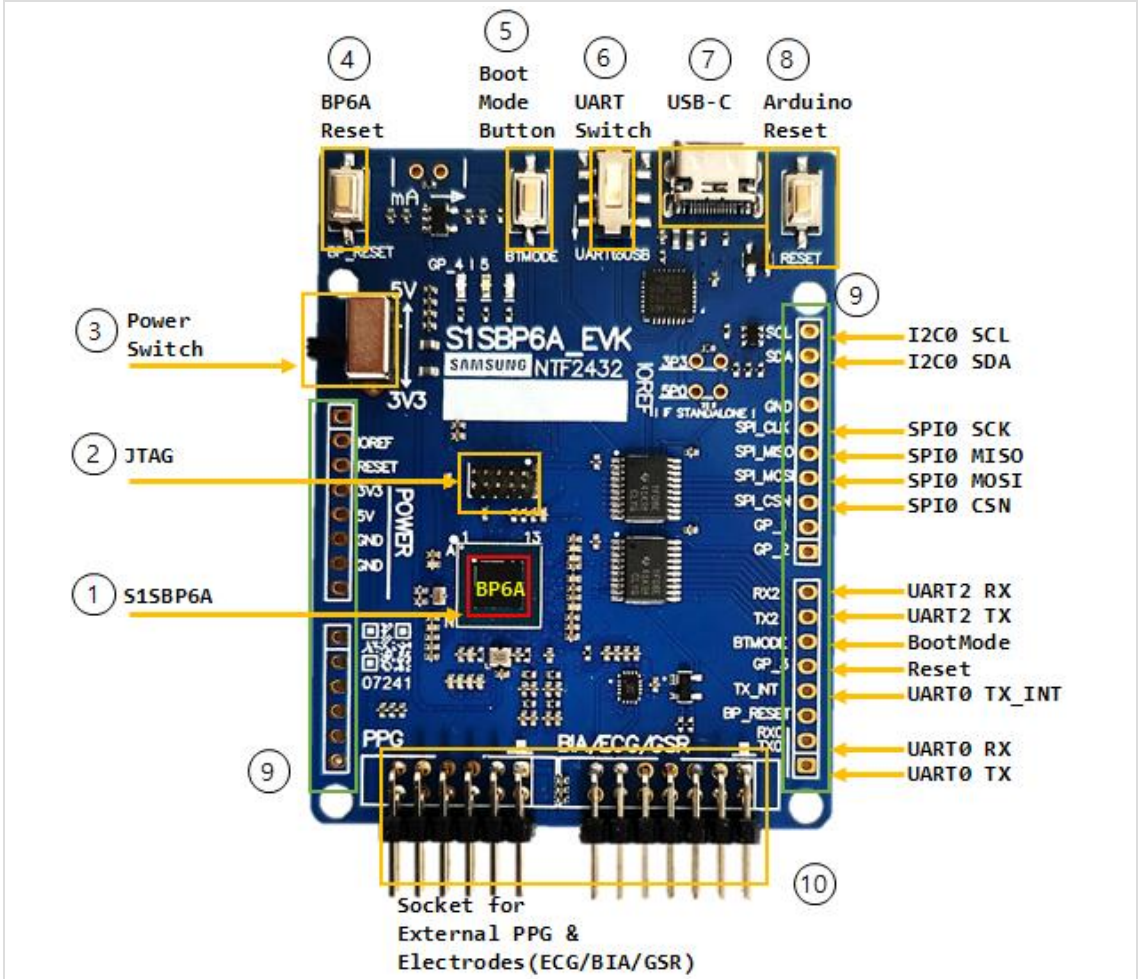
2.1 Components

2.1.1 Component Lists



Label	Name	Description
1	S1SBP6A_EVK Main Board	Evaluation board which has S1SBP6A bio-processor, Arduino compatible pins and sockets for electrode board.
2	Electrode Board	Electrode board which has dry electrodes for ECG/BIA/GSR and optical sensor for PPG.

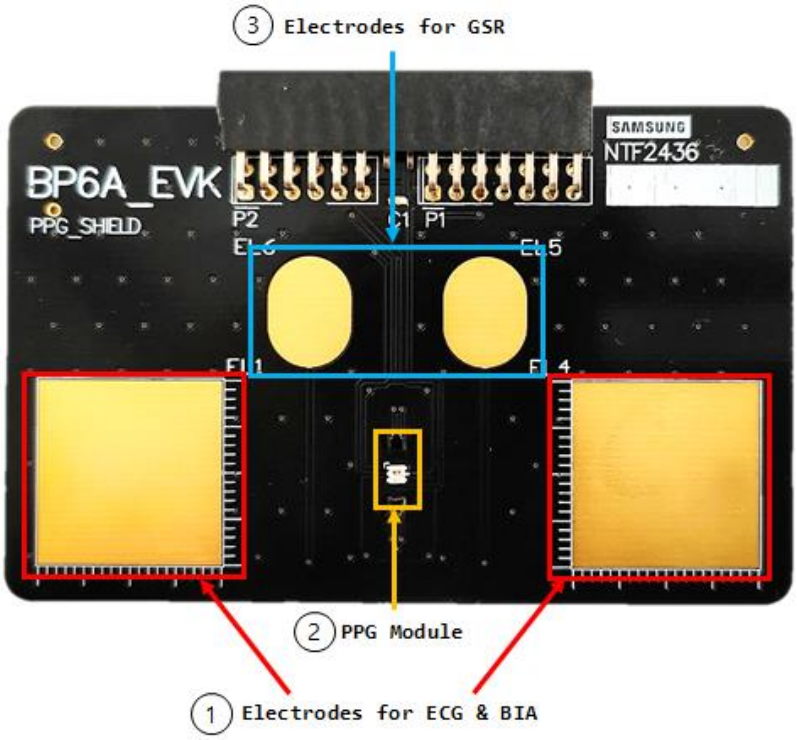
2.1.2 Main Device Components



Label	Name	Description
1	BP6A	S1SBP6A Samsung Bio-processor
2	JTAG	Socket for JTAG debugger
3	Power Switch	Switch for selecting Power0(USB 5V) or Power1(Arduino 3.3V)
4	BP_RESET	Button used when rebooting the BP6A
5	Boot Mode Button	Button used when entering boot mode (Operation sequence: While pressing button 5, press and release button 4, and then sequentially release button 5)
6	UARTtoUSB	Switch to transmit data to UART port of the Arduino or USB-C(w/ PC)
7	USB-C	USB-C for power supply and communication with PC
8	RESET	Button used when rebooting the Arduino
9	BP6A Pinout	Pinout compatible with Arduino Board

	I2C0 SCL	SCL(Serial Clock) of I2C0 in BP6A
	I2C0 SDA	SDA(Serial Data) of I2C0 in BP6A
	SPI0 SCK	Serial Clock of SPI0 in BP6A
	SPI0 MISO	Master Input Slave Output of SPI0 in BP6A
	SPI0 MOSI	Master Output Slave Input of SPI0 in BP6A
	SPI0 CSN	Chip Select of SPI0 in BP6A
	UART2 RX	RX(Receiver) of UART2 in BP6A
	UART2 TX	TX(Transmitter) of UART2 in BP6A
	BootMode	Low(0): BootRom Mode, High(1): Firmware Mode
	Reset	Low(0) to High(1): BP6A reboot
	UART0 TX_INT	gpio pin for interrupt used for UART0 TX control
	UART0 RX	RX(Receiver) of UART0 in BP6A
	UART0 TX	TX(Transmitter) of UART0 in BP6A
	IOREF	A voltage corresponding to the I/O of Arduino board
10	Socket for External PPG and Electrodes(ECG/BIA/GSR)	Socket used to extend external optical sensor modules for PPG and electrodes for ECG, BIA, and GSR

2.1.3 Accessory Components



Label	Name	Description
1	Electrodes for ECG & BIA	Embedded metal electrodes
2	PPG Module	Optical sensor module(SFH7072)
3	Electrodes for GSR	Embedded metal electrodes

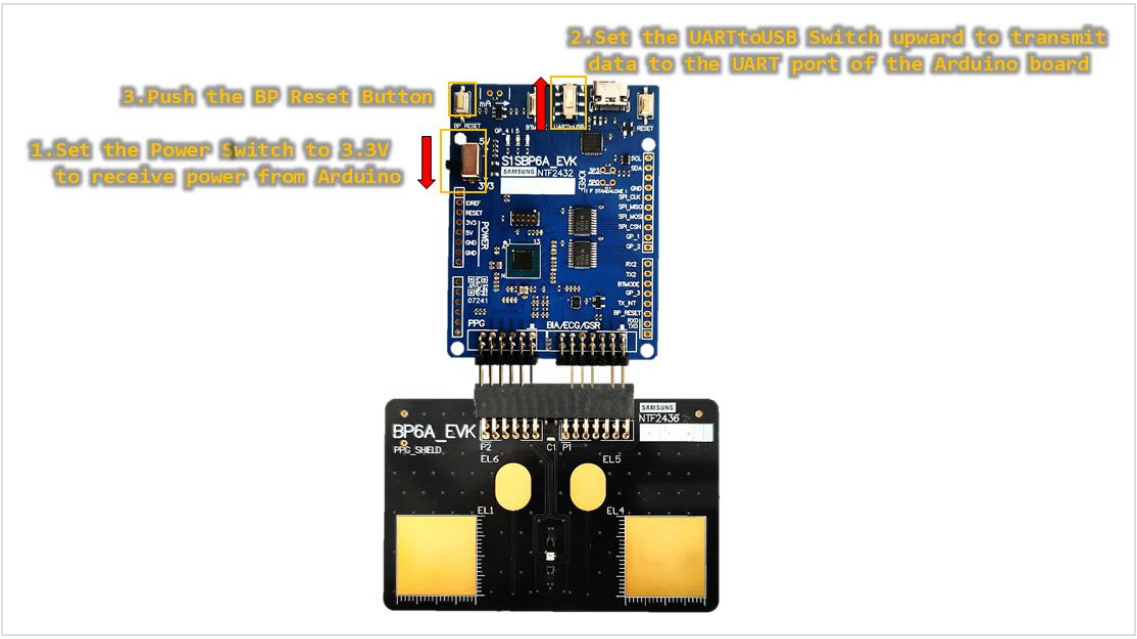
2.2 Preparation step for sensor measurement

2.2.1 Using electrode board

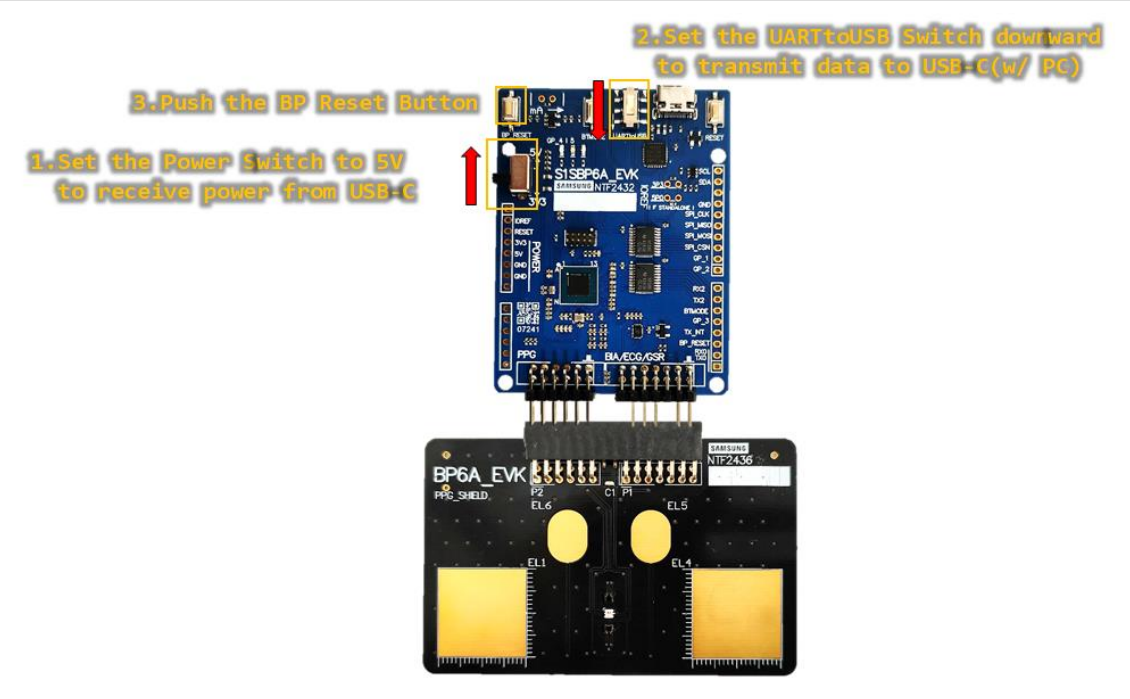
- Connect BP6A EVK Main Board and Electrode Board



2.2.2 Switch settings when connecting Arduino board



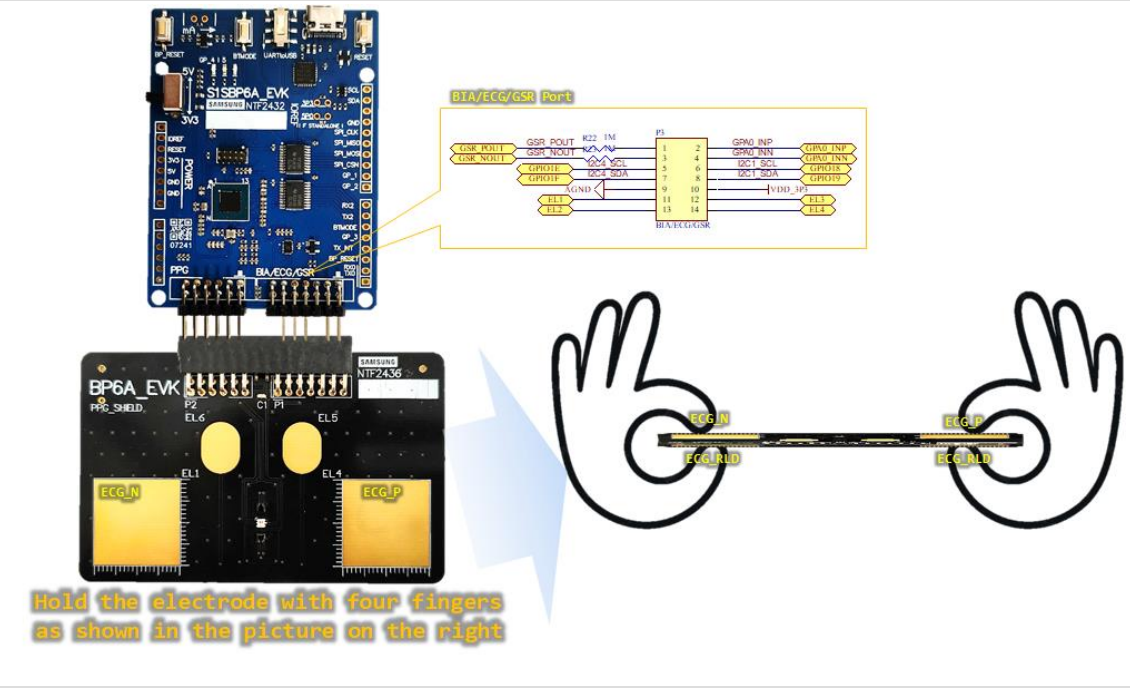
2.2.3 Switch settings when used alone without connecting Arduino board



2.3 How to set up hardware for ECG measurement

2.3.1 Using built-in electrodes and fingers

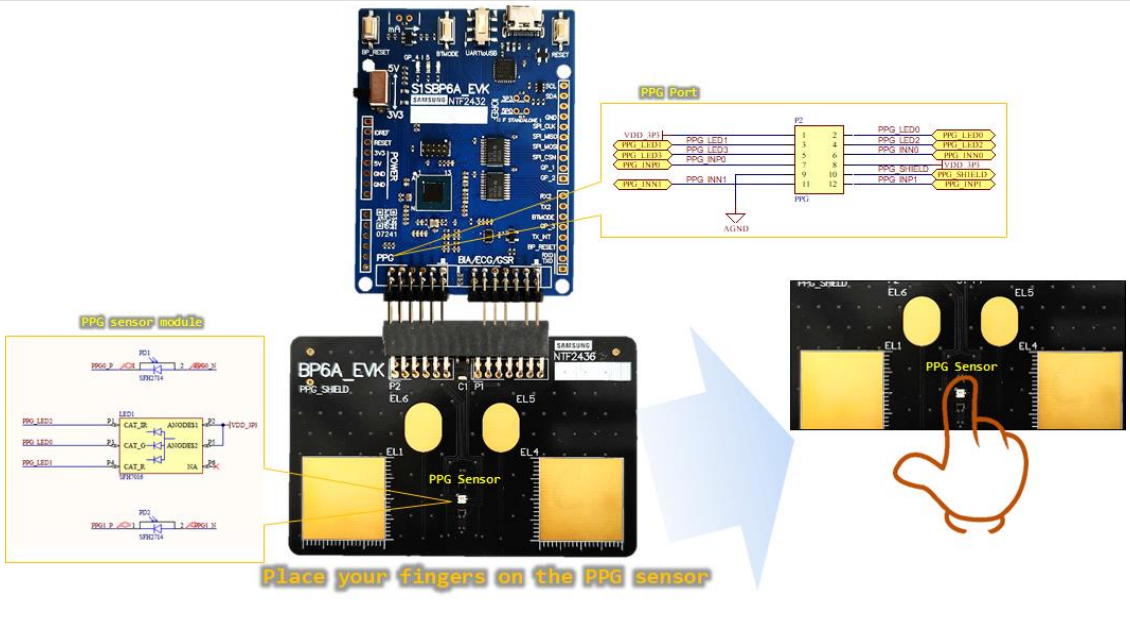
- Hold the electrode with four fingers as follows



2.4 How to set up hardware for PPG measurement

2.4.1 Using a built-in sensor module and your finger

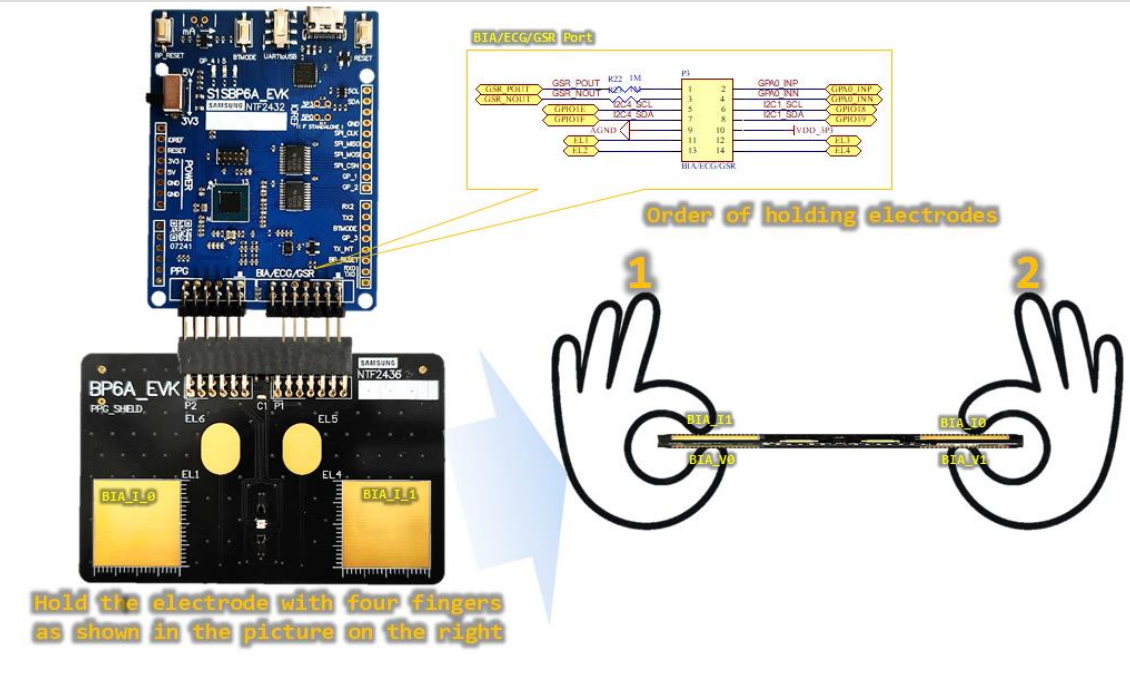
- Place your finger on the PPG sensor



2.5 How to set up hardware for BIA measurement

2.5.1 Using built-in electrodes and your fingers

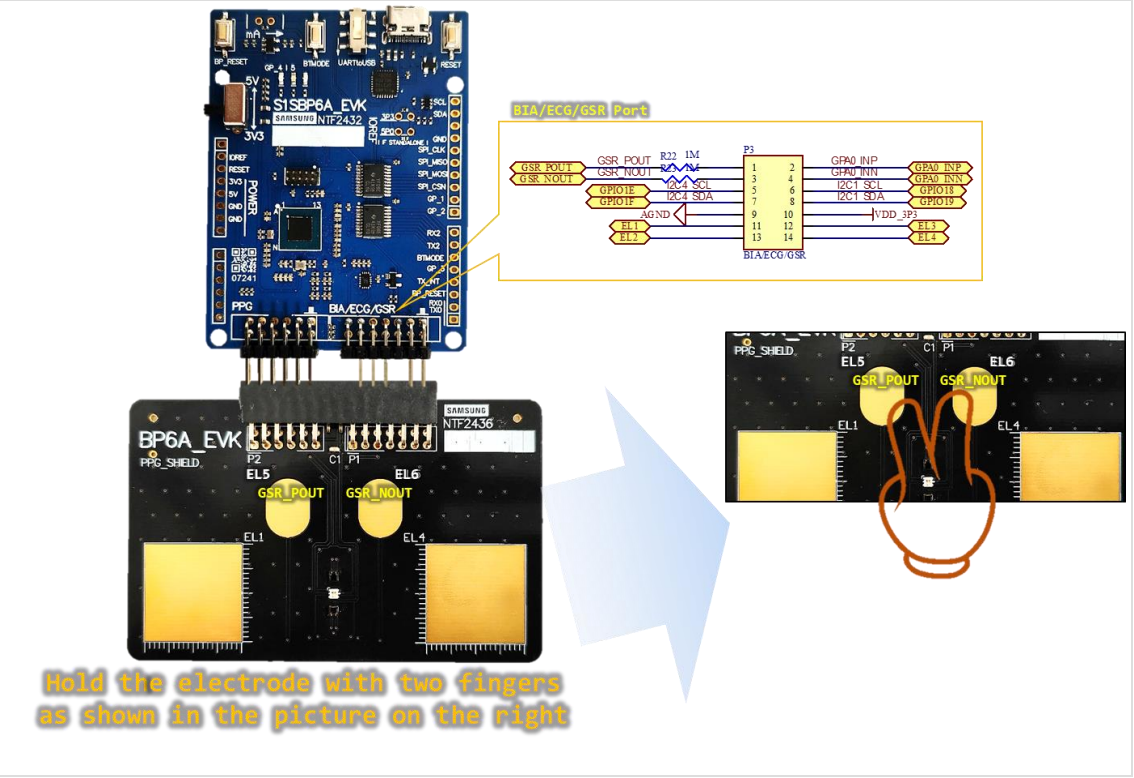
- Hold the electrode with four fingers as follows



2.6 How to set up hardware for GSR measurement

2.6.1 Using built-in electrodes and your fingers

- Hold the electrode with two fingers as follows



2.7 How to enter BOOT-ROM-MODE or NORMAL-MODE

2.7.1 Entering BOOT-ROM-MODE

- While holding down the BTMODE button, press and release the BP_RESET button.

After that, release the BTMODE button. BP6A will enter BOOT-ROM-MODE.



2.7.2 Entering NORMAL-MODE

- Press and release the BP_RESET button. BP6A will enter NORMAL-MODE.

